

Dihedral Angles, Scaled Jacobians, and a Signed Corner Energy for Hexahedral Finite Element Meshes

Notes compiled from a working session with Claude (Anthropic)

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Abstract

These notes collect, formalize, and derive the results developed in an interactive session on mesh-quality geometry. We first derive the dihedral angle between two triangular facets that share an edge. We then derive the scaled Jacobian (SJ) at a corner of a trilinear hexahedral element and show that it equals the signed volume of the parallelepiped spanned by the three unit edge vectors. We introduce an orthogonality-based *corner energy*, $E_{\cos} = \sum \cos^2 \theta_{ij}$, and show that it is blind to element inversion. We then construct a *signed corner energy* $E_{\pm} = \sum_k (1 - \hat{\mathbf{e}}_k \cdot \hat{\mathbf{n}}_{ij})$ that vanishes only at a perfect right-handed corner, equals 3 exactly at degeneracy, and keeps increasing as the corner inverts. All formulas are derived in full, the one-parameter test configuration used in the interactive illustrations is solved in closed form, and practical notes on embedding the illustrations in an mdBook are included. A record of the conversation is given in Appendix A.

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1 Notation

Vectors are bold ($\mathbf{a}$); unit vectors carry a hat, $\hat{\mathbf{a}} = \mathbf{a}/\|\mathbf{a}\|$. For two nonzero vectors, the (unsigned) angle between them is

$$\theta(\mathbf{a}, \mathbf{b}) = \arccos\left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}\right) \in [0, \pi]. \quad (1)$$

Two identities are used repeatedly:

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta(\mathbf{a}, \mathbf{b}), \quad (2)$$

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}) \quad (\text{Binet--Cauchy}). \quad (3)$$

The scalar triple product $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$ is invariant under cyclic permutation and changes sign under any transposition:

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{a}] = [\mathbf{c}, \mathbf{a}, \mathbf{b}] = -[\mathbf{b}, \mathbf{a}, \mathbf{c}]. \quad (4)$$

2 The dihedral angle between two triangular facets

2.1 Definition

Let two triangles $T_1 = (A, B, P)$ and $T_2 = (A, B, Q)$ share the edge AB , with apexes P and Q on (generally) different sides. The *dihedral angle* θ between them is the angle between the two half-planes bounded by line AB , measured in a plane perpendicular to AB .

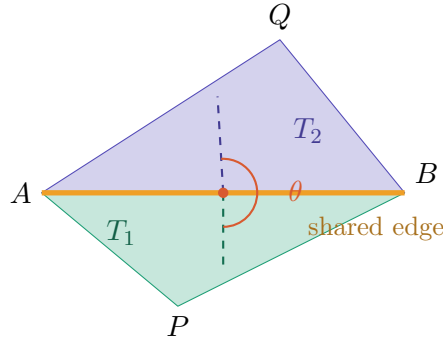


Figure 1: Two facets hinged on a shared edge. The dashed rays lie in each face and are perpendicular to the edge at a common point; the angle between them is the dihedral angle θ .

2.2 Derivation via perpendicular rays

Let $\hat{\mathbf{t}} = (\mathbf{B} - \mathbf{A})/\|\mathbf{B} - \mathbf{A}\|$ be the unit edge direction. Remove the component of each apex offset along the edge:

$$\mathbf{u} = (\mathbf{P} - \mathbf{A}) - ((\mathbf{P} - \mathbf{A}) \cdot \hat{\mathbf{t}})\hat{\mathbf{t}}, \quad \mathbf{v} = (\mathbf{Q} - \mathbf{A}) - ((\mathbf{Q} - \mathbf{A}) \cdot \hat{\mathbf{t}})\hat{\mathbf{t}}. \quad (5)$$

By construction $\mathbf{u} \perp \hat{\mathbf{t}}$ and $\mathbf{u}$ lies in the plane of T_1 (it is a linear combination of $\mathbf{P} - \mathbf{A}$ and $\mathbf{B} - \mathbf{A}$); likewise for $\mathbf{v}$ and T_2 . Hence $\mathbf{u}$ and $\mathbf{v}$ are exactly the perpendicular rays of Figure 1, and

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}. \quad (6)$$

2.3 Derivation via face normals

Define face normals with a consistent “from the edge toward the face” orientation: $\mathbf{n}_1 = \hat{\mathbf{t}} \times \mathbf{u}$ and $\mathbf{n}_2 = \hat{\mathbf{t}} \times \mathbf{v}$. By the Binet–Cauchy identity (3),

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = (\hat{\mathbf{t}} \cdot \hat{\mathbf{t}})(\mathbf{u} \cdot \mathbf{v}) - (\hat{\mathbf{t}} \cdot \mathbf{v})(\mathbf{u} \cdot \hat{\mathbf{t}}) = \mathbf{u} \cdot \mathbf{v}, \quad (7)$$

since $\hat{\mathbf{t}} \cdot \hat{\mathbf{t}} = 1$ and $\hat{\mathbf{t}} \cdot \mathbf{u} = \hat{\mathbf{t}} \cdot \mathbf{v} = 0$. Also $\|\mathbf{n}_1\| = \|\mathbf{u}\|$ and $\|\mathbf{n}_2\| = \|\mathbf{v}\|$ by (2), since $\mathbf{u} \perp \hat{\mathbf{t}}$. Therefore, with these normals, $\cos \theta = \hat{\mathbf{n}}_1 \cdot \hat{\mathbf{n}}_2$.

In a closed, consistently oriented surface the *outward* normals of two adjacent faces are $\mathbf{n}_1^{\text{out}} = \pm \mathbf{n}_1$ and $\mathbf{n}_2^{\text{out}} = \mp \mathbf{n}_2$ (one face is traversed in each direction across the shared edge). Hence

$$\cos \theta = - \frac{\mathbf{n}_1^{\text{out}} \cdot \mathbf{n}_2^{\text{out}}}{\|\mathbf{n}_1^{\text{out}}\| \|\mathbf{n}_2^{\text{out}}\|} \quad (\text{both outward or both inward}), \quad (8)$$

and the sign flips if one normal points in and the other out. Limiting cases: $\theta = \pi$ means the facets are coplanar (flat); $\theta \rightarrow 0$ means they fold shut.

2.4 Robust and signed evaluation

Because arccos is ill-conditioned near 0 and π , a numerically robust form is

$$\theta = \text{atan2}(\|\mathbf{u} \times \mathbf{v}\|, \mathbf{u} \cdot \mathbf{v}) \in [0, \pi]. \quad (9)$$

A *signed* dihedral angle in $(-\pi, \pi]$, distinguishing which way T_2 is rotated relative to T_1 about $\hat{\mathbf{t}}$, is

$$\theta_{\text{signed}} = \text{atan2}((\mathbf{u} \times \mathbf{v}) \cdot \hat{\mathbf{t}}, \mathbf{u} \cdot \mathbf{v}). \quad (10)$$

Remark 1. Viewing the configuration straight down the shared edge is the one viewpoint in which the on-screen angle equals θ , because that view is the plane perpendicular to $\hat{\mathbf{t}}$ seen head-on. This is the behavior demonstrated by the interactive illustration.

3 The scaled Jacobian at a hexahedral corner

3.1 From the trilinear map to edge vectors

A trilinear hexahedron maps the reference cube $[-1, 1]^3 \ni \boldsymbol{\xi} = (\xi, \eta, \zeta)$ to physical space by

$$\mathbf{x}(\boldsymbol{\xi}) = \sum_{a=1}^8 N_a(\boldsymbol{\xi}) \mathbf{x}_a, \quad N_a(\boldsymbol{\xi}) = \frac{1}{8}(1 + \xi_a \xi)(1 + \eta_a \eta)(1 + \zeta_a \zeta), \quad (11)$$

where $(\xi_a, \eta_a, \zeta_a) \in \{\pm 1\}^3$ are the reference coordinates of node a . The Jacobian matrix is $J = \partial \mathbf{x} / \partial \boldsymbol{\xi}$.

Lemma 1. *At a corner node a , the columns of J are one half of the three edge vectors leaving that node (oriented along increasing ξ, η, ζ reference directions).*

Proof. Take the corner $\boldsymbol{\xi}_0 = (-1, -1, -1)$ with node $\mathbf{x}_0$ and neighbors $\mathbf{x}_\xi, \mathbf{x}_\eta, \mathbf{x}_\zeta$ along the three reference axes. On the reference edge $\eta = \zeta = -1$ every shape function except those of $\mathbf{x}_0$ and $\mathbf{x}_\xi$ vanishes, and the map reduces to $\mathbf{x}(\xi, -1, -1) = \frac{1}{2}(1 - \xi)\mathbf{x}_0 + \frac{1}{2}(1 + \xi)\mathbf{x}_\xi$. Its derivative is $\partial \mathbf{x} / \partial \xi = \frac{1}{2}(\mathbf{x}_\xi - \mathbf{x}_0)$, and the same holds for η and ζ . Since the partial derivative along ξ evaluated at a point on that edge depends only on the restriction of $\mathbf{x}$ to the edge, these are the columns of $J(\boldsymbol{\xi}_0)$. Other corners follow by symmetry, with edge vectors oriented along the increasing reference directions. $\square$

Writing $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ for the three edge vectors at the corner (ordered so that the reference frame (ξ, η, ζ) is right-handed), Lemma 1 gives

$$\det J = \frac{1}{8} [\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]. \quad (12)$$

3.2 Scaling

Definition 1 (Corner scaled Jacobian).

$$\text{SJ} = \frac{[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]}{\|\mathbf{e}_1\| \|\mathbf{e}_2\| \|\mathbf{e}_3\|} = \hat{\mathbf{e}}_1 \cdot (\hat{\mathbf{e}}_2 \times \hat{\mathbf{e}}_3) = \det [\hat{\mathbf{e}}_1 \ \hat{\mathbf{e}}_2 \ \hat{\mathbf{e}}_3]. \quad (13)$$

The factor $\frac{1}{8}$ cancels against the same factor in the edge lengths of J 's columns, so (13) is the determinant of J with its columns normalized.

Geometric meaning. $[\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3]$ is the volume of the parallelepiped spanned by the three *unit* edges, and the sign records orientation: positive for a right-handed corner, negative for a left-handed (inverted) one.

Proposition 1 (Range). $-1 \leq \text{SJ} \leq 1$, with $\text{SJ} = 1$ if and only if the unit edges form a right-handed orthonormal triad, and $\text{SJ} = -1$ if and only if they form a left-handed one.

Proof. Write $\hat{\mathbf{n}} = \hat{\mathbf{e}}_2 \times \hat{\mathbf{e}}_3$. By (2), $\|\hat{\mathbf{n}}\| = \sin \theta_{23} \leq 1$, so by Cauchy–Schwarz $|\text{SJ}| = |\hat{\mathbf{e}}_1 \cdot \hat{\mathbf{n}}| \leq \|\hat{\mathbf{n}}\| \leq 1$ (Hadamard's inequality for this case). Equality $|\text{SJ}| = 1$ requires both $\sin \theta_{23} = 1$ (so $\hat{\mathbf{e}}_2 \perp \hat{\mathbf{e}}_3$) and $\hat{\mathbf{e}}_1 \parallel \hat{\mathbf{n}}$ (so $\hat{\mathbf{e}}_1 \perp \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$), i.e. an orthonormal triad; the sign of $\hat{\mathbf{e}}_1 \cdot \hat{\mathbf{n}}$ is its handedness. $\square$

Element minimum. The element's minimum scaled Jacobian is $\text{MSJ} = \min_{a=1..8} \text{SJ}_a$ over its eight corners (some implementations also include the element center). A single bad corner therefore sets the element's MSJ.

Regime	SJ	Geometry at the corner
Ideal	1	orthonormal, right-handed: a cube corner
Valid but distorted	$(0, 1)$	positive volume, skewed edges
Degenerate	0	the three edges are coplanar; zero volume; J singular
Inverted	< 0	left-handed; negative volume; element folded
Fully reflected	-1	orthonormal, left-handed

Table 1: Regimes of the corner scaled Jacobian.

4 The one-parameter test configuration

The interactive illustrations use a two-parameter family of corners. Place $\mathbf{e}_1$ and $\mathbf{e}_2$ in the xy -plane separated by the base angle $\varphi \in (0, \pi)$, and tilt $\mathbf{e}_3$ out of that plane by the elevation $\alpha \in [-\pi, \pi]$ while keeping its azimuth on the bisector $\beta = \varphi/2$:

$$\hat{\mathbf{e}}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{\mathbf{e}}_2 = \begin{pmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{pmatrix}, \quad \hat{\mathbf{e}}_3 = \begin{pmatrix} \cos \alpha \cos \beta \\ \cos \alpha \sin \beta \\ \sin \alpha \end{pmatrix}, \quad \beta = \frac{\varphi}{2}. \quad (14)$$

The perfect corner is $\varphi = \alpha = \pi/2$. For $\alpha \in (\pi/2, \pi]$ the edge $\hat{e}_3$ tips over the top and leans toward the opposite side; at $\alpha = \pm\pi$ it lies in the base plane, and at $\alpha = -\pi/2$ it points straight down. Throughout, write

$$s = \sin \alpha, \quad h = \cos \frac{\varphi}{2}, \quad \sigma = \sin \frac{\varphi}{2}. \quad (15)$$

4.1 Pairwise angles

$$\cos \theta_{12} = \hat{e}_1 \cdot \hat{e}_2 = \cos \varphi, \quad (16)$$

$$\cos \theta_{13} = \hat{e}_1 \cdot \hat{e}_3 = \cos \alpha \cos \beta = \cos \alpha h, \quad (17)$$

$$\cos \theta_{23} = \hat{e}_2 \cdot \hat{e}_3 = \cos \alpha (\cos \varphi \cos \beta + \sin \varphi \sin \beta) = \cos \alpha \cos(\varphi - \beta) = \cos \alpha h. \quad (18)$$

So placing $\hat{e}_3$ on the bisector makes $\theta_{13} = \theta_{23}$ for every α .

4.2 Scaled Jacobian

$\hat{e}_1 \times \hat{e}_2 = (0, 0, \sin \varphi)^\top$, so by cyclicity (4)

$$\boxed{\text{SJ}(\alpha, \varphi) = (\hat{e}_1 \times \hat{e}_2) \cdot \hat{e}_3 = \sin \varphi \sin \alpha = s \sin \varphi.} \quad (19)$$

SJ is the product of two factors: base skew ($\sin \varphi$) and out-of-plane tilt ($\sin \alpha$). It vanishes exactly when $\alpha \in \{0, \pm\pi\}$ (with $0 < \varphi < \pi$) and is negative exactly when $\alpha \in (-\pi, 0)$.

Remark 2 (Symmetry). Every quantity in this configuration depends on α only through $s = \sin \alpha$ and $\cos^2 \alpha = 1 - s^2$ (see (17)–(27)). All curves are therefore symmetric about $\alpha = \pi/2$ (and about $\alpha = -\pi/2$) on the full interval $[-\pi, \pi]$.

5 An orthogonality corner energy and its blind spot

Definition 2 (Unsigned corner energy).

$$\boxed{E_{\cos} = \cos^2 \theta_{12} + \cos^2 \theta_{13} + \cos^2 \theta_{23} = (\hat{e}_1 \cdot \hat{e}_2)^2 + (\hat{e}_1 \cdot \hat{e}_3)^2 + (\hat{e}_2 \cdot \hat{e}_3)^2.} \quad (20)$$

Each term is 0 exactly when that pair is at 90° and rises smoothly as the pair departs from 90° . Hence $E_{\cos} \geq 0$, $E_{\cos} = 0$ iff all three pairs are orthogonal, and $E_{\cos} \leq 3$ with equality iff all edges are parallel or antiparallel.

In the test configuration. From (16)–(18),

$$E_{\cos}(\alpha, \varphi) = \cos^2 \varphi + 2 \cos^2 \alpha \cos^2 \frac{\varphi}{2} = \cos^2 \varphi + 2(1 - s^2)h^2. \quad (21)$$

Proposition 2 (Inversion blindness). $E_{\cos}$ is invariant under $\hat{e}_k \mapsto -\hat{e}_k$ for any single edge, and more generally under any reflection of the triad. In particular $E_{\cos}$ cannot distinguish a perfect corner (SJ = 1) from a fully reflected one (SJ = -1): both have $E_{\cos} = 0$.

Proof. Each term $(\hat{e}_i \cdot \hat{e}_j)^2$ is even in each argument and invariant under orthogonal maps, including reflections. Reflecting $\hat{e}_3 \mapsto -\hat{e}_3$ in the perfect corner gives an orthonormal left-handed triad. $\square$

Along the test family with $\varphi = \pi/2$, $E_{\cos} = \cos^2 \alpha$: it rises from 0 at $\alpha = \pi/2$ to 1 at the degenerate corner $\alpha = 0$, then *falls back* to 0 as the corner inverts toward $\alpha = -\pi/2$. Over the full range $[-\pi, \pi]$ it has two equal minima, at $\pm\pi/2$, rating the perfect and the fully reflected corners as equally good.

Root cause. $\cos^2 \theta$ is an *even* function of the reflection $\psi \mapsto \pi - \psi$ of an edge across a plane. Any energy built only from unsigned angles inherits this blindness. To break the symmetry the energy must contain a quantity that is *odd* under reflection—i.e. it must involve orientation, which in three dimensions means a cross product (equivalently, the triple product that defines the Jacobian).

6 A signed corner energy

6.1 Definition

For each edge k , let (i, j, k) be the cyclic ordering $(1, 2, 3), (2, 3, 1), (3, 1, 2)$ and define the oriented unit normal of the plane of the other two edges,

$$\hat{\mathbf{n}}_{ij} = \frac{\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j}{\|\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j\|}, \quad (22)$$

and let $\psi_k \in [0, \pi]$ be the angle between $\hat{\mathbf{e}}_k$ and $\hat{\mathbf{n}}_{ij}$.

Definition 3 (Signed corner energy).

$$E_{\pm} = \sum_{k=1}^3 (1 - \cos \psi_k) = \sum_{(i,j,k) \text{ cyclic}} (1 - \hat{\mathbf{e}}_k \cdot \hat{\mathbf{n}}_{ij}). \quad (23)$$

The function $1 - \cos \psi$ is strictly increasing on $[0, \pi]$ and is *not* symmetric under $\psi \mapsto \pi - \psi$; this is precisely what restores orientation awareness. Equivalently, $1 - \cos \psi_k = 2 \sin^2(\psi_k/2)$.

6.2 Relation to the scaled Jacobian

Lemma 2. *For every cyclic (i, j, k) ,*

$$\hat{\mathbf{e}}_k \cdot \hat{\mathbf{n}}_{ij} = \frac{\text{SJ}}{\sin \theta_{ij}}, \quad (24)$$

and hence

$$E_{\pm} = \sum_{(i,j)} \left(1 - \frac{\text{SJ}}{\sin \theta_{ij}} \right) = 3 - \text{SJ} \left(\frac{1}{\sin \theta_{12}} + \frac{1}{\sin \theta_{23}} + \frac{1}{\sin \theta_{31}} \right). \quad (25)$$

Proof. By (2), $\|\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j\| = \sin \theta_{ij}$. By cyclicity (4), $\hat{\mathbf{e}}_k \cdot (\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j) = [\hat{\mathbf{e}}_k, \hat{\mathbf{e}}_i, \hat{\mathbf{e}}_j] = [\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3] = \text{SJ}$ for each cyclic (i, j, k) . Divide. $\square$

So $E_{\pm}$ is the scaled Jacobian reweighted by the inverse sines of the three face angles at the corner: it penalizes skew (small $\sin \theta_{ij}$) and inversion (negative SJ) through a single quantity.

6.3 Properties

Proposition 3. *Assume no two edges are parallel ($\sin \theta_{ij} > 0$). Then*

- (i) $0 \leq E_{\pm} \leq 6$;
- (ii) $E_{\pm} = 0$ if and only if the corner is perfect (orthonormal, right-handed, $\text{SJ} = 1$);
- (iii) $E_{\pm} = 3$ if and only if $\text{SJ} = 0$ (degenerate), independently of the face angles;

(iv) $E_{\pm} < 3$ iff $\text{SJ} > 0$ and $E_{\pm} > 3$ iff $\text{SJ} < 0$;

(v) $E_{\pm} = 6$ if and only if the corner is orthonormal and left-handed ($\text{SJ} = -1$).

Proof. (i) Each $\cos \psi_k \in [-1, 1]$, so each term lies in $[0, 2]$.

(ii) $E_{\pm} = 0$ iff every $\cos \psi_k = 1$, i.e. $\hat{\mathbf{e}}_k = \hat{\mathbf{n}}_{ij}$ for all k . Then $\hat{\mathbf{e}}_3 = \hat{\mathbf{n}}_{12} \perp \hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2$ and $\hat{\mathbf{e}}_1 = \hat{\mathbf{n}}_{23} \perp \hat{\mathbf{e}}_2$, so the triad is orthonormal, and $\text{SJ} = \hat{\mathbf{e}}_3 \cdot \hat{\mathbf{n}}_{12} \sin \theta_{12} = 1 > 0$. Conversely a right-handed orthonormal triad has $\hat{\mathbf{e}}_k = \hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j$ for every cyclic triple.

(iii)–(iv) From (25), $E_{\pm} - 3 = -\text{SJ} \sum_{(i,j)} 1/\sin \theta_{ij}$ and the sum is positive.

(v) Same as (ii) with $\hat{\mathbf{e}}_k = -\hat{\mathbf{n}}_{ij}$ for all k . $\square$

Property (iii) is what makes $E_{\pm}$ a natural companion to SJ : the degenerate threshold $\text{SJ} = 0$ maps to the fixed threshold $E_{\pm} = 3$ for every corner shape, and on each side the energy is ordered by orientation.

6.4 Closed form in the test configuration and monotonicity

In the configuration (14): $\sin \theta_{12} = \sin \varphi$, and from (17)–(18),

$$\sin \theta_{13} = \sin \theta_{23} = \sqrt{1 - h^2 \cos^2 \alpha} = \sqrt{1 - h^2(1 - s^2)} = \sqrt{\sigma^2 + h^2 s^2} \equiv \sqrt{D(s)}. \quad (26)$$

Substituting $\text{SJ} = s \sin \varphi$ into (25):

$$E_{\pm}(\alpha, \varphi) = 3 - \sin \alpha - \frac{2 \sin \varphi \sin \alpha}{\sqrt{\sin^2 \frac{\varphi}{2} + \cos^2 \frac{\varphi}{2} \sin^2 \alpha}} = 3 - s - \frac{2s \sin \varphi}{\sqrt{D(s)}}. \quad (27)$$

(The first term after 3 is the $\hat{\mathbf{e}}_3$ term: $\hat{\mathbf{n}}_{12} = \hat{\mathbf{z}}$, so $\cos \psi_3 = \sin \alpha$.)

Proposition 4 (Monotone through inversion). *For fixed $\varphi \in (0, \pi)$, $E_{\pm}$ is a strictly decreasing function of $s = \sin \alpha$ on $[-1, 1]$. Consequently, as $\hat{\mathbf{e}}_3$ tilts from vertical ($\alpha = \pi/2$) down through the plane ($\alpha = 0$) and into inversion ($\alpha \rightarrow -\pi/2$), $E_{\pm}$ increases strictly and continuously, with no turning point at $\text{SJ} = 0$.*

Proof. Let $f(s) = s/\sqrt{D(s)}$ with $D(s) = \sigma^2 + h^2 s^2 > 0$. Then

$$f'(s) = \frac{\sqrt{D} - s \cdot \frac{h^2 s}{\sqrt{D}}}{D} = \frac{D - h^2 s^2}{D^{3/2}} = \frac{\sigma^2}{D^{3/2}} > 0. \quad (28)$$

Hence

$$\frac{dE_{\pm}}{ds} = -1 - 2 \sin \varphi f'(s) = -1 - \frac{2 \sin \varphi \sin^2 \frac{\varphi}{2}}{D(s)^{3/2}} < 0. \quad (29)$$

Since $s = \sin \alpha$ is increasing on $[-\pi/2, \pi/2]$, $E_{\pm}$ is decreasing in α there, i.e. increasing as the tilt decreases. $\square$

Extremes. At $s = 1$, $D = 1$ and $E_{\pm} = 2 - 2 \sin \varphi$ (zero iff $\varphi = \pi/2$). At $s = -1$, $E_{\pm} = 4 + 2 \sin \varphi$ (six iff $\varphi = \pi/2$). Over the full sweep $\alpha \in [-\pi, \pi]$ the energy has a single minimum at $\alpha = \pi/2$ and a single maximum at $\alpha = -\pi/2$, and it passes through 3 exactly where $\text{SJ} = 0$ ($\alpha \in \{0, \pm\pi\}$).

φ	$E_{\pm}$ at $\alpha = 90^\circ$	$E_{\pm}$ at $\alpha = 0^\circ$	$E_{\pm}$ at $\alpha = -90^\circ$
90°	0	3	6
60°	$2 - \sqrt{3} \approx 0.268$	3	$4 + \sqrt{3} \approx 5.732$
20°	≈ 1.316	3	≈ 4.684

Table 2: Signed energy at the perfect-tilt, degenerate, and fully-inverted tilts.

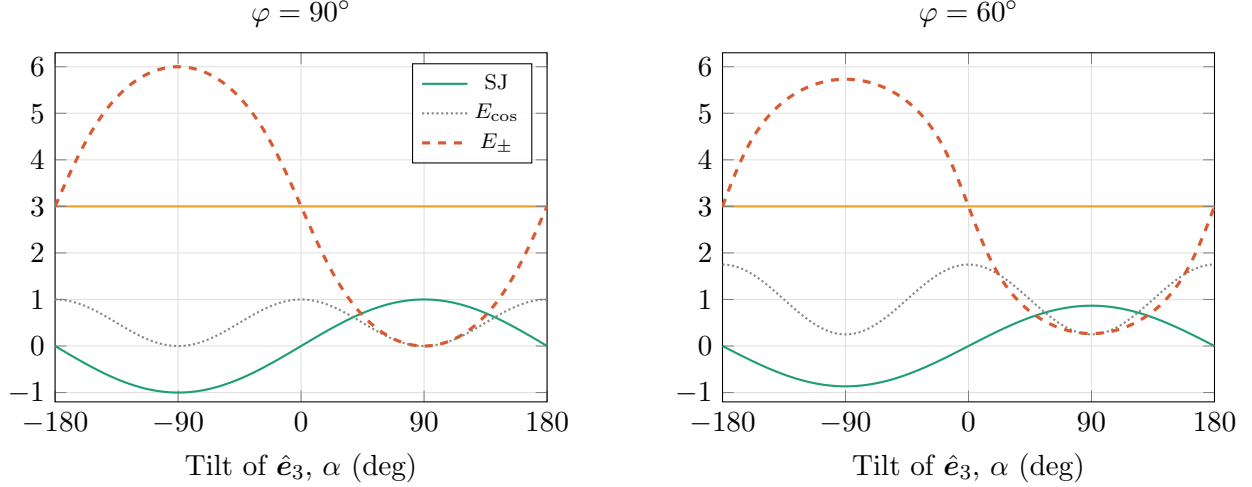


Figure 2: Scaled Jacobian, unsigned energy $E_{\cos}$, and signed energy $E_{\pm}$ along the tilt sweep, for a square base (left) and a skewed base (right). $E_{\pm}$ crosses the line 3 exactly where SJ crosses 0 and keeps rising through inversion; $E_{\cos}$ turns back down and has a spurious second minimum at the fully reflected corner $\alpha = -90^\circ$.

6.5 A simpler alternative and why it needs tuning

One may instead keep $E_{\cos}$ and add an orientation penalty,

$$E_w = E_{\cos} + w(1 - \text{SJ}), \quad w > 0. \quad (30)$$

$E_w \geq 0$ with equality only at the perfect corner (both terms must vanish, and $\text{SJ} = 1$ forces orthonormal right-handedness by Proposition 1). But monotonicity through inversion depends on w . In the test configuration, using (19) and (21),

$$E_w = \cos^2 \varphi + 2(1 - s^2)h^2 + w(1 - s \sin \varphi), \quad \frac{dE_w}{ds} = -4h^2s - w \sin \varphi. \quad (31)$$

This is negative for all $s \in [-1, 1]$ iff it is negative at the worst case $s = -1$:

$$4h^2 < w \sin \varphi \iff w > \frac{4 \cos^2 \frac{\varphi}{2}}{2 \sin \frac{\varphi}{2} \cos \frac{\varphi}{2}} \iff \boxed{w > 2 \cot \frac{\varphi}{2}}. \quad (32)$$

For a square base this needs $w > 2$; for $\varphi = 60^\circ$, $w > 2\sqrt{3} \approx 3.46$; for $\varphi = 20^\circ$, $w \gtrsim 11.3$. The required weight grows without bound as the base collapses, whereas $E_{\pm}$ is monotone for every φ with no parameter to tune (Proposition 4).

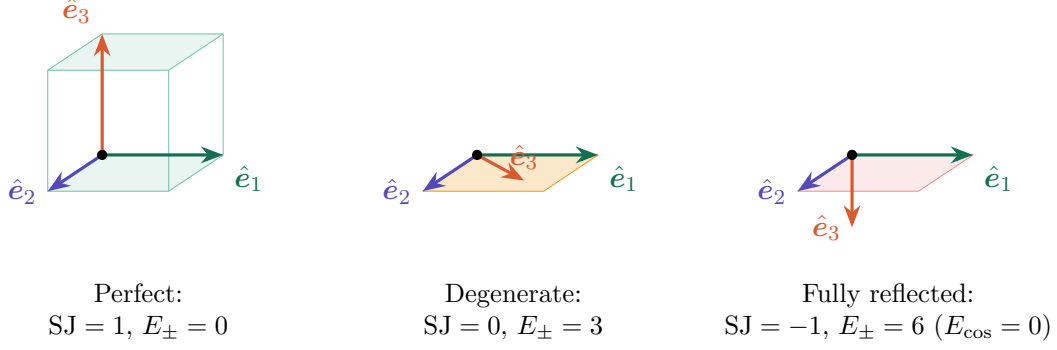


Figure 3: Three corners of the test family with a square base. The unsigned energy $E_{\cos}$ is zero for both the left and right corners; the signed energy separates them.

6.6 Practical notes

- **Parallel edges.** $\hat{n}_{ij}$ is undefined when $\sin \theta_{ij} \rightarrow 0$. Clamp $\|\hat{e}_i \times \hat{e}_j\| \geq \varepsilon$ in code. In an optimizer, it can help to keep a $\cos^2$ term alongside $E_{\pm}$, since $\cos^2 \theta_{ij}$ remains smooth there.
- **Barriers versus finite penalties.** A barrier such as $-\log SJ$ forbids inversion outright (it is infinite at $SJ = 0$). $E_{\pm}$ is finite everywhere, which is what one wants when an inverted mesh must be *untangled* by descent: the gradient keeps pointing back toward validity on the inverted side.
- **Element energy.** An element energy can be formed as $\sum_{a=1}^8 E_{\pm}^{(a)}$, or as $\max_a E_{\pm}^{(a)}$ to mirror the MSJ.

6.7 Reference implementation

```
import numpy as np

CYCLIC = [(0, 1, 2), (1, 2, 0), (2, 0, 1)]

def corner_metrics(e1, e2, e3, eps=1e-12):
    """Return (SJ, E_cos, E_signed) for three edge vectors at a corner node.

    Edges must be ordered so that the reference frame is right-handed.
    """
    e = [np.asarray(v, float) / np.linalg.norm(v) for v in (e1, e2, e3)]
    sj = float(np.dot(np.cross(e[0], e[1]), e[2]))
    e_cos = sum(float(np.dot(e[i], e[j]))**2 for i, j in [(0, 1), (0, 2), (1, 2)])
    e_signed = 0.0
    for i, j, k in CYCLIC:
        n = np.cross(e[i], e[j])
        e_signed += 1.0 - float(np.dot(e[k], n)) / max(np.linalg.norm(n), eps)
    return sj, e_cos, e_signed
```

The closed forms (19), (21), and (27) agree with this vector implementation to machine precision ($\leq 10^{-15}$) over a grid of $\alpha \in [-180^\circ, 180^\circ]$, $\varphi \in [20^\circ, 160^\circ]$.

7 Embedding interactive illustrations in an mdBook

mdBook passes raw HTML through to the rendered page, so inline `<svg>` and `<script>` work in any chapter. Three approaches, from simplest to most robust:

1. **Inline HTML in the chapter.** Works, but the Markdown parser ends an HTML block at the first blank line, so the snippet—including the script—must contain no empty lines.
2. **Markup in the chapter, code in a separate file (recommended).** Keep only a container in the chapter and register the script and stylesheet in `book.toml`. The script locates every widget by class, so one widget can appear in several chapters or several times on a page.
3. **An `<iframe>` to a standalone HTML page.** Fully isolated from mdBook’s styles, but the height must be fixed by hand and it will not follow the reader’s theme.

Chapter snippet (option 2)

```
<div class="dihedral-widget">
<svg viewBox="0 0 680 400" width="100%"><g class="scene"></g></svg>
<input type="range" min="10" max="180" value="110">
</div>
```

book.toml

```
[output.html]
additional-js = ["widgets/dihedral.js"]
additional-css = ["widgets/widgets.css"]
```

When porting widgets written for another host, replace host-specific CSS classes and variables (e.g. `var(--fg)` for text, `var(--bg)` for background) so the widget follows the Light, Rust, Coal, Navy, and Ayu themes. Hard-coded data colors (faces, edges, arcs) can stay as they are.

A Record of the conversation

The user’s requests are quoted verbatim; the assistant’s responses are summarized.

1. **User:** “illustrate the dihedral angle between two triangular facets that share an edge”

Assistant: Produced an interactive 3D illustration: two triangular faces hinged on an amber shared edge, dashed rays in each face perpendicular to the edge at a common point, a coral arc marking θ , a slider for $\theta \in [10^\circ, 180^\circ]$, and drag-to-rotate. Explained that viewing down the edge shows the true angle, and gave the normal-based formula (Section 2).

2. **User:** “wow, this is a very nice interactive illustration. Is there a way to include such a nice interactive illustration in an mdbook?”

Assistant: Explained the three embedding options (inline HTML with no blank lines; container markup plus `additional-js/additional-css` in `book.toml`; an `<iframe>`), recommended the second, and noted replacing host styles with mdBook theme variables (Section 7).

3. **User:** “is there a second illustration you can create, of minimum Scaled Jacobian for the the corner of a hexahedron of a finite element mesh with three incident edges at a single node? I would love to be able to illustrate the MSJ going to zero and going negative, and how the perfect MSJ of 1.0 looks.”

Assistant: Built an interactive hex corner: three edge vectors from a node, the unit parallelepiped they span (colored green/amber/red for valid/degenerate/inverted), a live SJ readout and a $[-1, 1]$ gauge, sliders for the tilt of $\hat{e}_3$ and the angle between $\hat{e}_1$ and $\hat{e}_2$, and presets (perfect, skewed, degenerate, inverted). Gave $SJ = \sin \varphi \sin \alpha$ for the configuration and described the three regimes (Sections 3–4).

4. **User:** “is there a way to show ‘corner energy’ as a B panel (A versus B panel) to the hex, where the corner energy is zero for a perfect 90-90-90 degree corner, and then increases as the angle between two axes separates from 90”

Assistant: Added panel B plotting SJ and $E_{\cos} = \sum \cos^2 \theta_{ij}$ against the tilt, next to the 3D corner in panel A. Pointed out that $E_{\cos}$ is blind to inversion: it falls back to 0 for a fully reflected corner (Section 5).

5. **User:** “is there a way to embed the sign of the corner in the corner energy such that the corner energy continues to increase when the MSJ goes from zero to negative?”

Assistant: Explained that $\cos^2$ is even under reflection, so orientation must enter through the oriented normal. Introduced $E_{\pm} = \sum_k (1 - \hat{e}_k \cdot \hat{n}_{ij}) = \sum (1 - SJ / \sin \theta_{ij})$, with $E_{\pm} = 0$ only at the perfect corner, $E_{\pm} = 3$ exactly at $SJ = 0$, and $E_{\pm} > 3$ when inverted. Updated panel B with all three curves. Noted the parallel-edge guard and the alternative $E_{\cos} + w(1 - SJ)$, which needs $w \geq 2 \cot(\varphi/2)$ (Section 6).

6. **User:** “For the x-axis ‘Tilt of e_3’, it current runs from -60 deg to 90 deg. Can we make that axis run from -180 degrees to 180 degrees?”

Assistant: Extended the slider and axis to $[-180^\circ, 180^\circ]$ and the energy axis to $[-1, 6]$, added a “fully reflected” preset at -90° , and reported separate θ_{13}, θ_{23} and ψ_3 . Over the full sweep, SJ is a sine wave; $E_{\cos}$ has two equal minima at $\pm 90^\circ$; $E_{\pm}$ has a single minimum at $+90^\circ$, a single maximum at -90° , and crosses 3 exactly where $SJ = 0$ (Figure 2).

7. **User:** “can you write down this entire conversation, along with a formulation, into a LaTeX .tex document I can download, please include all the derivations of the formulae.”

Assistant: Produced this document.